1) i) Append the Required zeros

Original Data Bits = 110001011

Generator Bits : 1001 (n = 4)

number of zero to append (n-1) = 3

Augmented Data = 110001011000

ii)

```
            1101010
     1001 ) 110001011000
            1001
             1010
             1001
              01110
                  0
              01110
                1001
                01110
                 1001
                 0111
                  1001
                  01110
                   1001
                    100
```

iii) CRC = 100

iv) 110001011100

2, i, $G(x) = x^3 + x + 1 \Rightarrow x^3 + x^2 + x^1 + x^0$

$\Rightarrow 1011$

ii, Append zeros

message = 11000

n = 3

= 11000000

iii)

```
1011 | 11000000
       1011
       -----
        01110
         1011
        -----
         01010
          1011
         -----
          00010
```

$\Rightarrow 010$

iv, $C_1 = 0$

$C_2 = 1$

$C_3 = 0$

v, codeword = 11000010

Q.3 $d8 = 1$ $d7 = 1$ $d6 = 0$ $d5 = x$ $d4 = 0$ $d3 = 1$ $d2 = 0$ $d1 = 1$

1, $C_8 = y$ $C_4 = 0$ $C_2 = 1$ $C_1 = 0$

$x = 0 \quad y = 0$

Parity bit C_1 check positions

1, 3, 5, 7, 9, 11

values , 0, 1, 0, 0, x, 1

Parity bit check C_8

8, 9, 10, 11, 12

values: y, 0, 0, 1, 1

$y = 0$

C_2 check

position:

2, 3, 6, 7, 10, 11

values :

1, 1, 1, 0, 0, ~~0~~ 1

C_4 check

4, 5, 6, 7, 12

0, 0, 1, 0, 1

3. Final code word:

110000100110

4. 0000

No error

4) $A = 00000$

$B = 01011$

$C = 10101$

$D = 11110$

1) $HD(A, B) = 3$

$HD(A, C) = 3$

$HD(A, D) = 4$

$HD(B, C) = 4$

$HD(B, D) = 3$

$HD(C, D) = 3$

2) $P = 3$

3) $q = \left[\frac{P-1}{2} \right]$

$q = \left[\frac{3-1}{2} \right] = 1$

$q = 1$ bit.